

David Dangerfield

Principal Software Engineer

Consumer Products | Health | Payments | CleanTech | Enterprise Software | 20+ Years

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SUMMARY

Principal Software Engineer with a passion for the craft of coding, testing, shipping, and supporting software across the full stack to delight clients. Still in the code daily: shipping fixes, upgrades, and new features across large systems with dozens of repos. Solid foundational education (MS in CS, Purdue) and decades of deep-system engineering are now enhanced by **using modern agentic AI workflows to deliver rapid solutions** without sacrificing quality. Broad industry experience across backend and frontend, from operating system modules to UI components. Track record includes FDA-regulated healthcare platforms, billion-dollar revenue systems, and products serving millions of users. Independent consultant since 2021, delivering principal-level engineering to select engagements. Open to full-time roles for the right fit. A collaborative engineer who engages with teammates to learn, share ideas, and foster trust.

TECHNICAL PROFILE

Languages: TypeScript, JavaScript, Java, Kotlin, Python

Backend & Frameworks: REST APIs, Node.js, Spring Boot, Legacy Modernization, Microservices, Message Queues

State Mgmt & Data Stores: TanStack, Pinia, Redux, MySQL, MongoDB, PostgreSQL

Cloud & Infrastructure: AWS, GitHub Actions, Azure DevOps, Docker, Kubernetes

AI & Testing Tools: Copilot, Claude, Playwright, Cypress, Jest, JUnit

Frontend: Vue.js, React, Next.js, GraphQL, Angular, Tailwind, ECharts, a11y, i18n, Design Systems, Component Libraries

EXPERIENCE

Principal Software Engineer (Contractor)

Mar 2025 - Aug 2026

[Patterson Companies \(Remote\)](#)

- **Led frontend architecture** for a B2B client portal with associated admin experience, consolidating duplicated UI and Vue.js routing into a shared repository with Storybook and mock data
- **Designed an app launch pattern** enabling decoupled app initialization while preserving shared code
- **Upgraded a multibillion-dollar revenue stream** by rewriting legacy desktop-only Angular.js bill-pay frontend into a responsive, locale-aware Vue.js/TypeScript/TanStack architecture
- **Transitioned into agentic AI development** to improve my dev efficiency and software quality
- **Engineered a reusable UI component library** from Figma designs, integrating Vue.js with a .NET/C# controller backend and mapping SAP responses to user feedback
- **Built a localized Customer Reports experience** integrating embedded Power BI reports across e-commerce sites, including the supporting UI and integration architecture
- **Spearheaded multi-repo infrastructure modernizations**, including an ESLint9 upgrade and an org-wide JFrog Artifactory migration, both across 22 repositories

Tech: Copilot, Claude, Node.js, TypeScript, Vue.js, Microfrontends, Microservices, Azure DevOps, JFrog, i18n

Lead Software Developer (Contractor)

Jul 2022 - Mar 2025

[Life Time \(Remote\)](#)

- **Led frontend development** for the Account team, digitizing the member guest pass experience to support hundreds of thousands of new accounts and app installs
- **Spearheaded Vue 3 implementation** for "Find A Location" featuring custom Google Maps integration
- **Replaced legacy JavaScript libraries** in core platform code to eliminate technical debt

Tech: GitHub Copilot, TypeScript, Vue.js, Bootstrap, Vite, Vitest, Playwright, Legacy Modernization, App Integration

Senior Software Engineer (Concurrent Part-Time Contractor)

Feb 2018 - Mar 2024

[Nurocor \(Remote\)](#)

- **Architected compliant UI workflows** for a structured authoring platform for FDA clinical trial documentation, supporting document templates, formal reviews, and complete audit trails
- **Delivered award-winning frontend engineering** for a protocol authoring platform recognized as "Best Integrated Protocol Authoring and Review Solution" by TransCelerate BioPharma

Tech: Vue.js, Nuxt.js, Storybook, GraphQL, Component Libraries, Kubernetes

Principal Software Engineer Consultant (Contractor)

Jan 2021 - Jul 2022

NextEra Analytics (Remote)

- **Developed "OMNI" project UIs** featuring geospatial maps, diagrams, and dense financial data grids
- **Introduced Apache ECharts to unlock** a broad library of interactive, animated visualizations not previously practical to build in-house, enabling new product and data analytics charts
- **Engineered an Angular-based iPad app** via web views to streamline wind-site worker scheduling

Tech: TypeScript, Vue.js, GraphQL, ECharts, Angular, Storybook, Python, Terraform, AWS

Senior Software Consultant (Concurrent Part-Time Contractor)

Sep 2020 - Feb 2022

Intricately (Acq. by HG Insights) (Remote)

- **Engaged directly by CTO** as a frontend expert to provide technical oversight and establish best practices across a more junior international team
- **Mentored developers and built trust** through code reviews and Vue 2 to Vue 3 migration plans

Tech: TypeScript, Vue.js, Node.js

Senior Software Engineer

Sep 2018 - Jan 2021

SmugMug (Remote)

- **Migrated legacy payments code** to Stripe, ensuring a locale-aware UI across 10 global currencies
- **Engineered responsive trial-to-subscriber conversion flows** supporting dynamic promotions
- **Implemented data-driven conversion A/B testing** via LaunchDarkly, integrating detailed tracking analytics and user feedback mechanisms, collaborating closely with product, UX, and design

Tech: AWS, Node.js, JavaScript, React, Tailwind, GraphQL, Continuous Deployment, Legacy Modernization, a11y, i18n

Advisory Software Engineer with Watson Health & IBM Services

Nov 2015 - Aug 2018

IBM (Rochester, MN)

- **Led frontend engineering** for Watson for Drug Discovery (Pfizer), iterating with UX on complex D3.js visualizations to translate 25M+ abstracts and 4M+ patent records into actionable insights
- **Built Vue.js application** for "Project DOC," a \$3M funded clinical decision support application deployed at Su Clinica Familiar to optimize regional diabetes treatment models
- **Served as sole UI developer** for a custom Vue.js app for call center staff at a major regional U.S. bank; delivered in under 2 months, with time onsite in Alabama, enabling NLP-ingested document search
- **Co-founded FED@IBM Rochester**, an internal frontend developer community of 30+ engineers hosting twice-monthly sessions on JavaScript, accessibility, UI automation testing, and tooling

Tech: IBM Bluemix (Cloud), Vue.js, Angular, React, JavaScript, i18n, D3.js, Mobile Web App, Legacy Modernization, a11y

Senior Web Application Developer

Aug 2013 - Oct 2015

Preventice Solutions (now Boston Scientific) (Rochester, MN)

- **Built full-stack infrastructure** for BodyGuardian, an FDA 510(k)-cleared ambulatory cardiac monitoring platform incorporating Mayo Clinic licensed remote monitoring technology
- **Engineered an ActiveMQ message broker pipeline** for the Notification Service to batch and deliver structured ambulatory telemetry reports to clinical review teams in a fault-tolerant manner
- **Built Java web services** for patient compliance app with integration test suite and code coverage
- **Implemented claims-based (SAML) authentication flows** for the multi-tenant Preventice Care Platform, including multi-role user sessions, ensuring appropriate HIPAA-compliant access controls

Tech: Java, JavaScript, C#, REST APIs, ActiveMQ, SAML SSO, ASP.NET, SQL Server, Angular

Software Engineer → Team Lead → Test Architect

May 2002 - Aug 2013

IBM (Rochester, MN)

- **Ported IBM Directory Server (LDAP)** to IBM i, a commercially shipped enterprise product
- **Co-designed test strategy** and tool ecosystem for Blue Business Platform, a new product line, leading engineering teams in Beijing to run product validation across multiple simultaneous releases
- **Automated software update packaging**, eliminating 500+ person-hours annually and enabling self-service builds across teams

Tech: C++, LDAP, REST APIs, Java, JavaScript, Perl, IBM Db2, i18n

EDUCATION

Master of Science in Computer Science | Purdue University, 2002

Bachelor of Science in Computer Science | Brigham Young University, 2000